Who Said What? Speaker Attribution in a Conversational Power Benchmark

Muttaqi Ahmad Alladin[*] Purdue University

Krish Patel[*] Purdue University

Jiwon Kim[*] Purdue University

Abstract

Wu et al. (2025) ask which of two speakers in a six-turn meeting excerpt is the professor, and report that PhD judges, GPT-3.5, and an XOR factor graph over LLM-written arguments all score near 50%. We trace all 151 released test snippets back to the ICSI Meeting Corpus and find that their displayed speaker letters and their labels disagree: under the reading of the letters that makes the labels correct, 57.6% of the words are shown under the other speaker's letter. We rebuild the snippets with speakers taken from the corpus. On the corrected speech, a decision model (Jev) identifies the professor in 71.3% of 150 snippets against a 56.7% majority baseline, where it scored 48.0% on the released text, and it reaches 89.3% with 60 segments of surrounding meeting. With the data rebuilt we revisit the method. The factor graph, rerun call by call with two current LLMs, reaches 72.0% and 69.3%, against 72.7% and 73.3% for asking those LLMs directly, and neither rebuttals, relation factors, nor Dung's grounded extension improve on a direct answer. Code, data and every model output are released, and one command regenerates the data and all reported numbers.

1 Introduction

Can a model tell who holds power in a conversation from a few turns? Wu et al. (2025) study this with professor-student exchanges of six turns from the ICSI Meeting Corpus (Janin et al., 2003), taking the professor to be the higher-power speaker. They report 46.3% accuracy for six PhD judges, 49.0% for GPT-3.5 on the excerpt alone, and 54.3% for their method, a factor graph over LLM-written reasons and rebuttals. The picture is of a task that is close to impossible at this length, on which structured argumentation gives a small gain.

This project began as an attempt to improve that method and to beat the majority-class baseline, which none of the published systems clearly exceeds. We proposed quote-checked arguments and Dung's grounded extension (Dung, 1995) in place of the factor graph, and found that neither helped and that nothing we tried on the released test set beat the majority label. We analyzed potential causes for the poor accuracy by tracing the released snippets back to the corpus, and adjusted our proposal by adding a System One decision model. To summarize, our contributions are three-fold:

- **Correcting and adjusting the test set.** In the released test file of Wu et al. (2025) the displayed speakers and the labels disagree (Section 3). Under the reading of the speaker letters that makes the labels correct, 57.6% of the words are shown under the other speaker's letter. We rebuild the 150 usable snippets from the corpus, at the original span and with longer context, with a builder that regenerates them from public sources (Section 4).
- **The task is feasible.** On the rebuilt snippets a decision model and three LLMs identify the professor in 71 to 76% of cases from the same speech, and accuracy rises to 89% with more of the surrounding meeting (Section 6).
- **The argument graph does not add accuracy.** With clean data, the factor graph, its rebuttals and relations, and the grounded extension do not improve on a direct answer for any of three argument-writing models (Section 6.3).

2 Background

The method of Wu et al. For each of four aspects (style, content, coordination, engagement) an LLM (GPT-3.5) summarizes the conversation, writes three reasons why A has more power and three why B does, and writes a rebuttal to each reason. The same LLM then evaluates each reason and each rebuttal for validity and grounding through its

[*]These authors contributed equally.

yes/no token probabilities, following the method of Kassner et al. (2023) for scoring statements.

Once the reasons, rebuttals and scores are in hand, a factor graph is built over them. The graph lets the strength of one reason bear on another, in contrast to verifying each reason independently: a reason and its rebuttal cannot both hold, a speaker has power exactly when one of their reasons holds, and exactly one speaker has power. The graph is solved with the AD3 algorithm (Martins et al., 2011), and the speaker it selects is the prediction of who has more power in the dialogue. The paper suggests abstract argumentation (Dung, 1995) as a later formalization.

Related work. Earlier work predicts power from word choice, accommodation and thread structure with supervised models (Danescu-Niculescu-Mizil et al., 2012; Prabhakaran and Rambow, 2014), and Ziems et al. (2024) find that zero-shot LLMs often trail such models on social science tasks, power among them. Sanayei et al. (2025) find that LLMs only moderately match argumentation-theoretic semantics when asked to judge debate graphs. Freedman et al. (2025) compute decisions from LLM-written arguments with gradual semantics and report accuracy comparable to chain-of-thought prompting, and Smit et al. (2024) find that multi-agent debate does not reliably beat self-consistency or ensembling.

3 The released test set

Sources. The released test set is the file `meeting.csv` of Wu et al. (2025): 151 rows, each with a conversation between speakers A and B and a label naming the higher-power speaker.[1] Speaker A is the label in 83 rows, so always answering A scores an accuracy of 55.0%. The snippets come from the ICSI Meeting Corpus. We use its original MRT transcripts, in which every segment carries a start time, a participant ID and the transcribed text, and its participant metadata, which records each participant's education level.[2]

Locating the snippets. After removing transcript markup, the non-empty lines of a released snippet are a run of consecutive corpus segments. Matching on that sequence finds all 151 snippets, each in exactly one place, across 45 meetings. This step uses only the released text. It gives, for every line of every snippet, the participant who said it.

Shown	Said by	Text
A	professor	Uh, maybe you have to…
A	other	Mm-hmm.
B	professor	but -
A	other	Yeah, I guess.
B	professor	Yeah.
A	professor	do you have any way of…
A	other	Mmm.
B	other	No, we don't.
B	professor	Yeah.

Table 1: Released snippet 0 (meeting Bro025, starting at 1273.5 s, abridged). "Shown" is the letter the line appears under in the released text. "Said by" is from the corpus participant IDs. The released label is B.

Displayed speakers against corpus speakers. In the released text, each displayed turn ends with the first segment of the next corpus speaker. In Table 1 the turn shown under A contains the professor's remarks and then the other participant's "Mm-hmm." A letter therefore does not stand for one participant throughout a snippet, and there are two ways to read it.

If each letter is read as the speaker of the last line of its turns, the letters map one-to-one onto the two participants in 128 snippets with exactly one professor, and the released label names the professor in 128 of those 128. Under that reading, 7,596 of the 13,185 words in those snippets (57.6%) are displayed under the other participant's letter, and in 73 of the 128 snippets more than half of the words are.

If instead each letter is read as whoever says most of the words shown under it, the text is mostly attributed correctly, and the released label names the letter that carries most of the professor's words in only 69 of the 128 snippets.

Under either reading the text and the labels disagree: the letter that a reader sees doing the professor's talking is, in about half of the snippets, not the letter the label rewards. In the remaining 23 snippets a letter covers two participants (22) or only one participant speaks (1). Appendix A lists every snippet.

What we can and cannot conclude. The cached GPT-3.5 arguments in the release were generated from this text: the conversation stored with each

[1] https://github.com/kandicew/social-reasoning, commit fed11b6, data/meeting.csv.

[2] https://groups.inf.ed.ac.uk/ami/icsi/download: ICSI_original_transcripts.zip and ICSI_core_NXT.zip (speakers.xml), distributed under CC BY 4.0.

cached snippet is identical to the corresponding row of `meeting.csv` for all 151. The code that built the snippets from the corpus is not in the public repository, so we cannot say which step produced the mismatch. We do not know whether the human judges of Wu et al. (2025) saw the same text, or whether the numbers in their paper were computed from this file. We have not yet confirmed these observations with the authors.

Consequence. A reader or model that judges the displayed text will, in many snippets, credit the professor's behaviour to the other letter. On the released file that is what we observe: the rule "whoever says more words is the professor" scores 46.0%, the direct Jev question 48.0%, and no method we ran exceeds the majority rate (Appendix B). It also predicts that the same judgments become accurate once speech is attributed to the people who said it.

4 The rebuilt test set

We rebuilt each snippet from the located corpus segments. Speakers are the corpus participant IDs, consecutive segments by one participant form a turn, and the professor is the participant whose education level in `speakers.xml` is Professor. A snippet is kept if it has exactly one professor and at least two speakers, which drops one snippet and leaves 150. In the 128 snippets above, the letters follow the last-line reading, so the released labels carry over unchanged; in the other 22, the first of the two speakers to talk is A. In total 142 of the 150 labels equal the released label, and the professor is A in 85 (a 56.7% base rate). Three snippets include a third participant, shown as C.

Each snippet is rendered at three lengths: its own segments (the same words as the released snippet), and 15 or 60 further corpus segments on each side, with other participants shown as C, D, and so on. The other speaker's education level is PhD in 117 snippets, Grad in 30 and Postdoc in 3, so "student" follows the usage of Wu et al. (2025). The builder is deterministic and regenerates the set byte for byte from the three source files.

5 Methods

Direct questions. Jev is a decision model that returns a probability for each typed yes/no question about a supplied state.[3] We ask "Is speaker X the professor?" for A and for B, and predict the speaker with the higher probability as the professor. We also ask four LLMs which speaker has more power as baselines: GPT-6.1 Terra, GLM-5.3, Claude Haiku and Claude Sonnet. Additionally, we use two predefined rules that need no model: (1) the speaker with more words is the professor, and (2) the speaker with longer turns is the professor.

The factor graph of Wu et al. We ran their method on the rebuilt snippets with three sets of arguments. For the first, GPT-6.1 Terra wrote the arguments call by call with the prompt templates of Wu et al. (2025): one independent request for each aspect summary, each list of three reasons, and each rebuttal (36 requests per snippet). Terra performed these actions with tools disabled and no access to the labels. GLM-5.3 wrote the second set through the same call-by-call pipeline. For the third, Claude Haiku wrote all arguments for a snippet in one pass. Jev scored all three sets: validity and grounding of each argument, whether each rebuttal counters its reason, and similarity and contradiction of reason pairs. A separate Haiku instance also scored the Haiku arguments. Their rule file and weights (reason 20, rebuttal 18) do the inference, solved exactly as an integer program, in the three variants of the paper: reasons only, with rebuttals, and with all relations. We solve the graph exactly in place of the approximate AD3 solver, and the same solver reproduces the published 54.3% on the released cache (82 of 151).

Typed-statement factor graph. For this method, we keep the same factor graph and replace the generated text. For each aspect we wrote three yes/no statements, such as "Does speaker X evaluate speaker Y's work?". Jev's probability for a statement about X is the weight of a reason for X, and the same statement about the other speaker is its rebuttal. One call with 24 questions replaces the generation and scoring calls for 48 arguments per snippet.

Grounded extension. Using the same 24 reasons and 24 rebuttals, we replace the factor graph with Dung's grounded labelling. It is a framework that draws conclusions from arguments that attack one another, and it assigns each argument one of three labels:

- An argument is IN when every argument at-

[3] TypeSafe AI, `jev-1.14.0`, `https://docs.typesafe.ai`. The questions used on the rebuilt snippets were fixed before that data was built.

tacking it is OUT. An argument with no attackers is therefore IN.

- An argument is OUT when at least one of its attackers is IN.
- Every other argument is undecided (UNDEC).

After the labelling, the speaker with more IN reasons is predicted as the one with more power. A reason has strength $P(\text{valid}) \cdot P(\text{grounded})$ and a rebuttal $P(\text{counters its reason}) \cdot P(\text{grounded})$. In the relation of our proposal every rebuttal attacks its reason and opposing reasons attack each other. In a preference-based relation an attack succeeds only if the attacker is at least as strong as its target. Appendix C gives details.

Gradual argumentation. As a soft alternative to the labelling, we follow Freedman et al. (2025) and build a quantitative bipolar argumentation framework for the claim "A has more power": reasons for A support it, reasons for B attack it, and each rebuttal attacks its reason, with the same strengths as base scores. The claim's strength is computed with the DF-QuAD or the Quadratic Energy semantics, over a flat graph or one grouped by aspect, from a prior of 0.5 or of the direct Jev answer, and A is predicted when it exceeds 0.5. All eight variants are reported.

Debate with a judge. Following the debate protocol of Khan et al. (2024), GLM-5.3 reads the conversation together with its own twelve reasons for each speaker, each followed by its rebuttal, and then decides. Every snippet is judged twice, with A's case first and with B's case first.

Statistics. Paired comparisons use the exact McNemar test. Intervals come from a bootstrap that resamples whole meetings.

6 Results

6.1 Attribution

Table 2 compares the released and rebuilt inputs. On the rebuilt snippets at the original span, the direct Jev question identifies the professor in 107 of 150 cases. Against always answering A this is 34 wins and 12 losses ($p = 0.0016$), and the meeting-level bootstrap gives a 95% interval of 6.7 to 22.0 points for the difference. The same question on the released text scores 72 of 150, and the paired contrast between the two inputs is 52 wins to 17 losses ($p < 0.0001$). The words are the same in both conditions; only who is shown saying them differs. The word-count rules move the same way, from 46.0% to 66.7%, though their paired tests against the majority label on the rebuilt set are not significant ($p = 0.10$ and $p = 0.053$).

	Rel.	Rebuilt	±15	±60
Always A	55.3	56.7	56.7	56.7
More words	46.0	66.7	66.0	68.7
Longer turns		68.0	69.3	66.7
Jev, professor	48.0	71.3	84.7	89.3
Jev, more power	48.0	70.7	79.3	88.0
Typed-statement graph	51.3	70.0	80.7	90.7

Table 2: Accuracy (%) on the 150 snippets. "Rel." uses the released text and labels for the same snippets. "Rebuilt" uses the same words with speakers from the corpus. The last two columns add 15 or 60 corpus segments per side. The blank cell was not run.

Args.	Scores	Dir.	Reas.	+Reb.	All	Gr.
Terra	Jev	72.7	66.7	72.0	72.0	70.0
GLM	Jev	73.3	71.3	70.7	69.3	67.3
Haiku	Jev	68.0	73.3	70.0	70.0	73.3
Haiku	Haiku	68.0	68.7	67.3	68.0	64.7

Table 3: Argument-based methods on the rebuilt snippets at the original span (accuracy, %). Rows give who wrote and who scored the arguments. "Dir." is the argument-writing model asked directly. "Reas.", "+Reb." and "All" are the three variants of the factor graph of Wu et al. (2025). "Gr." is the grounded extension with the preference relation. For comparison, the direct Jev question scores 71.3% and Claude Sonnet asked directly scores 76.0%.

The word-count rules do not improve with context, so the gain is not explained by the professor simply talking more. GLM-5.3 asked directly gains less: 73.3%, 74.0% and 82.0% at the three lengths, below Jev at ±15 (9 wins to 25 losses, $p = 0.009$) and not distinguishable from it at ±60 (10 to 21, $p = 0.07$).

Jev's probabilities also become informative. Its Brier score is 0.189 at the original span against 0.246 for the base rate, and its most confident fifth of answers is 93.3% correct.

6.2 Context

More context helps. With 15 extra segments per side Jev reaches 84.7% (30 wins to 10 losses against the original span, $p = 0.002$), and with 60 it reaches 89.3% with a Brier score of 0.095.

6.3 Does the argument graph help?

Table 3 gives the argument-based methods. With Terra's arguments and Jev's scores the full factor graph reaches 72.0%. That is level with Terra asked directly (72.7%, 14 wins to 15 losses) and with the direct Jev question (14 to 13). With GLM's arguments it reaches 69.3%, against 73.3% for GLM asked directly (18 wins to 24 losses, $p = 0.44$). With Haiku's arguments it reaches 70.0% under Jev's scores and 68.0% under Haiku's, against 68.0% for Haiku asked directly.

The parts of the graph do not behave consistently across the sets of arguments. Rebuttals add 5.3 points with Terra's arguments (11 wins to 3 losses, $p = 0.057$), cost 3.3 points with Haiku's under the same scorer, and cost 0.7 with GLM's. Under Jev's scores, adding the relations changes nothing with Terra's or Haiku's arguments and costs 1.3 points with GLM's. A plain sum of reason scores with no graph equals the reasons-only graph in all four rows. The typed-statement graph tracks the direct Jev question at every context length (Table 2; 21 wins to 23 losses at the original span, $p = 0.88$).

The grounded extension tells the same story. Under the relation of our proposal no reason is ever IN, so every decision is made by the tie-break. The preference relation gives a real decision on every snippet with Jev scores, at 70.0% with Terra's arguments, 67.3% with GLM's and 73.3% with Haiku's. A search over 288 grounded policies, with the policy chosen on half of the meetings and scored on the other half, averages 69.7% against 71.5% for the direct question on the same halves, and 67.1% over GLM's arguments (Appendix C).

Gradual semantics do not change this. Over the eight variants the claim-strength decision scores 68.7 to 72.7% with Terra's arguments, 69.3 to 72.0% with GLM's and 70.0 to 74.0% with Haiku's, and none of the 24 results differs from the direct Jev question (smallest $p = 0.34$).

Letting the model read the arguments does not help either. As a judge of its own arguments GLM-5.3 scores 67.3% with A's case first and 69.3% with B's case first, against 73.3% when asked directly (18 wins to 27 losses, $p = 0.23$, and 22 to 28, $p = 0.48$). It gives the same answer in both orders for 117 of 150 snippets and is correct in 86 of those; order matters little, but the judge answers A in 215 of 300 judgments. Taking the direct answer when the two orders disagree gives 75.3% (17 to 14, $p = 0.72$).

Combining judgments, in contrast to combining arguments, may help: a majority vote of the four direct answers (GLM, Terra, Sonnet and Jev, ties to Jev) scores 77.3%, against 71.3% for Jev alone (15 wins to 6 losses, $p = 0.078$).

The best single result at the original span is a stronger model asked directly (Sonnet, 76.0%), which is not distinguishable from Jev (27 wins to 20 losses, $p = 0.38$) or from Terra (15 to 10, $p = 0.42$); GLM asked directly is also level with Jev (24 to 21, $p = 0.77$). We find no evidence that structured argumentation over generated reasons adds accuracy on this task. With 150 snippets a difference of a few points would not be detectable.

6.4 Robustness

The rebuilt set has four professors, and one of them is the professor in 110 of the 150 snippets. On the 40 snippets with the other three professors, Jev scores 77.5%, 85.0% and 85.0% at the three context lengths, Terra 77.5% and Sonnet 70.0%, so the result does not depend on the most frequent professor. None of these models is trained on the task.

The first of the two speakers to talk is the professor in 83 of 150 snippets, so turn order alone is no better than the majority label.

Jev favours the letter A at the original span: it is correct in 85.9% of the snippets where the professor is A and in 52.3% where the professor is B. Its balanced accuracy is therefore 69.1%, a little below the 71.3% headline, and its AUC is 0.79. The asymmetry disappears with context (89.4% and 89.2% at ± 60). Sonnet shows the opposite, smaller asymmetry (71.8% and 81.5%).

7 Limitations

The order of work was exploratory. The mismatch in the released file was found, and the rebuilt set constructed, after many runs on the released file, which has no development split. The questions used on the rebuilt snippets were written before that data existed, and each run was made once.

Our observations about the released file have not been confirmed with its authors, and we cannot tell from public materials whether the experiments in Wu et al. (2025) used the released text. We did not run a human study, so we cannot say how people perform on the rebuilt snippets.

No rerun of the factor graph uses the original model. The Terra and GLM runs follow the

pipeline call by call, but the text comes from GPT-6.1 Terra or GLM-5.3 and the scores from Jev, because neither interface we used exposes token probabilities. GLM-5.3 reasons before every answer, and its direct answer is the letter it prints; one of its 3,600 rebuttals came back empty and was scored as it was. The Haiku run wrote each snippet's arguments in one pass, and its Haiku scores are stated probabilities with about 15 distinct values. We did not rerun GPT-3.5 or the quote-filter experiments on the rebuilt snippets.

The test set is small and comes from one corpus of research meetings with four professors. With 60 extra segments a model can use cues that a six-turn excerpt lacks, including how others address the two speakers. Jev is a proprietary model in early access whose training and calibration claims are not peer reviewed; the LLM baselines give similar accuracy at the original span.

8 Reproducibility

The repository contains the rebuilt snippets with corpus offsets, every model answer, and all code.[4] One command downloads the five external inputs (the released `meeting.csv`, argument cache and rule file, and the two ICSI annotation archives), verifies each by SHA-256, regenerates the snippet set byte for byte, and recomputes every number in this report from the stored model answers without calling a model. The model answers themselves come from hosted models and cannot be regenerated exactly; the scripts and instructions used to collect them are included.

References

Cristian Danescu-Niculescu-Mizil, Lillian Lee, Bo Pang, and Jon Kleinberg. 2012. Echoes of power: Language effects and power differences in social interaction. In *Proceedings of the 21st International Conference on World Wide Web*, pages 699–708.

Phan Minh Dung. 1995. On the acceptability of arguments and its fundamental role in nonmonotonic reasoning, logic programming and n-person games. *Artificial Intelligence*, 77(2):321–357.

Gabriel Freedman, Adam Dejl, Deniz Gorur, Xiang Yin, Antonio Rago, and Francesca Toni. 2025. Argumentative large language models for explainable and contestable claim verification. In *Proceedings of the AAAI Conference on Artificial Intelligence*.

Adam Janin, Don Baron, Jane Edwards, Dan Ellis, David Gelbart, Nelson Morgan, Barbara Peskin, Thilo Pfau, Elizabeth Shriberg, Andreas Stolcke, and Chuck Wooters. 2003. The ICSI meeting corpus. In *Proceedings of ICASSP*, volume 1, pages I–364.

Nora Kassner, Oyvind Tafjord, Ashish Sabharwal, Kyle Richardson, Hinrich Schütze, and Peter Clark. 2023. Language models with rationality. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 14190–14201, Singapore. Association for Computational Linguistics.

Akbir Khan, John Hughes, Dan Valentine, Laura Ruis, Kshitij Sachan, Ansh Radhakrishnan, Edward Grefenstette, Samuel R. Bowman, Tim Rocktäschel, and Ethan Perez. 2024. Debating with more persuasive LLMs leads to more truthful answers. In *Proceedings of the 41st International Conference on Machine Learning*.

André F. T. Martins, Mário A. T. Figueiredo, Pedro M. Q. Aguiar, Noah A. Smith, and Eric P. Xing. 2011. An augmented Lagrangian approach to constrained MAP inference. In *Proceedings of the 28th International Conference on Machine Learning*, pages 169–176.

Vinodkumar Prabhakaran and Owen Rambow. 2014. Predicting power relations between participants in written dialog from a single thread. In *Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 339–344, Baltimore, Maryland. Association for Computational Linguistics.

Reza Sanayei, Srdjan Vesic, Eduardo Blanco, and Mihai Surdeanu. 2025. Can LLMs judge debates? evaluating non-linear reasoning via argumentation theory semantics. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 21244–21262. Association for Computational Linguistics.

Andries Petrus Smit, Nathan Grinsztajn, Paul Duckworth, Thomas D. Barrett, and Arnu Pretorius. 2024. Should we be going MAD? A look at multi-agent debate strategies for LLMs. In *Proceedings of the 41st International Conference on Machine Learning*.

Zhaoqing Wu, Dan Goldwasser, Maria Leonor Pacheco, and Leora Morgenstern. 2025. Identifying power relations in conversations using multi-agent social reasoning. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers)*, pages 855–865, Albuquerque, New Mexico. Association for Computational Linguistics.

Caleb Ziems, William Held, Omar Shaikh, Jiaao Chen, Zhehao Zhang, and Diyi Yang. 2024. Can large language models transform computational social science? *Computational Linguistics*, 50(1):237–291.

[4] https://github.com/muttaqiahmadalladin/cs-577

A Differences between the released file and the rebuilt set, by snippet

Table 4 has one row for each of the 151 released snippets. **ID** is the row index in the released `meeting.csv`. **Meeting** and **Start** give the ICSI meeting and the start time in seconds of the snippet's first corpus segment. **Seg.** and **Words** are the numbers of corpus segments and words in the snippet. **Misplaced** is the number of words displayed under the other participant's letter when each released letter is read as the speaker of the last line of its turns, and **%** is that number as a share of the snippet's words; both are blank when the letters do not map one-to-one onto two participants. **Rel.** is the released label and **Reb.** is the professor's letter in the rebuilt set, marked † where the two differ. **Letters** says whether the rebuilt letters follow the released ones (kept) or the first of the two speakers to talk was made A (reassigned).

In total, 150 of the 151 snippets are in the rebuilt set. Letters are kept in 128 and reassigned in 22, and the label differs from the released label in 8. Over the 128 snippets with a one-to-one mapping, 7,596 of 13,185 words are misplaced; 121 of them have at least one misplaced word, 7 have none, and the median share is 55%. The same rows are in `data/released_vs_rebuilt.csv` in the repository.

Table 4: Every released snippet against the rebuilt set.

ID	Meeting	Start	Seg.	Words	Misplaced	%	Rel.	Reb.	Letters	Note
0	Bro025	1273.5	15	98	89	91	B	B	kept	
1	Bro008	332.0	11	218	113	52	B	B	kept	
2	Bro003	618.0	10	50	31	62	A	A	kept	
3	Bmr029	1138.4	14	122	–	–	B	A†	reassigned	a letter covers two participants
4	Bed004	3006.4	10	143	90	63	B	B	kept	
5	Bro005	2065.7	10	71	35	49	B	B	kept	
6	Bro027	3074.1	10	83	17	20	A	A	kept	
7	Bmr015	1350.5	15	117	66	56	B	B	kept	
8	Bro013	1135.9	8	49	20	41	A	A	kept	
9	Bmr001	1839.5	8	87	37	43	B	B	kept	
10	Bed011	1007.8	12	108	66	61	A	A	kept	
11	Bro011	1266.0	69	356	–	–	A	A	reassigned	a letter covers two participants
12	Bed004	1404.9	8	80	26	32	A	A	kept	
13	Bmr025	613.7	9	83	38	46	A	A	kept	
14	Bmr006	748.7	8	73	3	4	A	A	kept	
15	Bmr009	579.0	12	103	–	–	A	B†	reassigned	a letter covers two participants
16	Bmr014	493.0	11	138	83	60	B	B	kept	
17	Bmr029	789.9	6	59	0	0	A	A	kept	
18	Bro008	777.2	27	472	321	68	A	A	kept	
19	Bmr005	2826.9	8	64	12	19	A	A	kept	
20	Bmr001	1916.4	8	55	15	27	B	B	kept	
21	Bro004	1902.2	11	81	49	60	A	A	kept	
22	Bro026	2882.0	10	88	29	33	A	A	kept	
23	Bro027	999.1	13	91	50	55	B	B	kept	
24	Bro017	1268.5	6	44	0	0	A	A	kept	
25	Bro005	1215.7	6	31	–	–	A	A	reassigned	a letter covers two participants
26	Bmr029	885.3	8	54	23	43	B	B	kept	
27	Bro027	2509.2	6	41	0	0	A	A	kept	
28	Bmr006	2931.2	8	87	–	–	B	B	reassigned	a letter covers two participants
29	Bmr008	371.2	8	37	15	41	A	A	kept	
30	Bro008	4.3	16	136	83	61	A	A	kept	
31	Bro005	2429.2	19	107	75	70	A	A	kept	
32	Bro007	577.3	12	71	55	77	A	A	kept	
33	Bro027	3401.9	8	69	3	4	B	B	kept	
34	Bed014	1023.1	8	39	5	13	B	B	kept	
35	Bro017	1982.4	14	183	114	62	B	B	kept	
36	Bmr006	651.8	10	86	45	52	A	A	kept	third participant
37	Bro023	1211.8	21	273	218	80	A	A	kept	
38	Bro017	2372.0	13	84	44	52	B	B	kept	
39	Bro005	3410.5	22	249	–	–	A	A	reassigned	a letter covers two participants
40	Bro017	1881.0	9	123	27	22	B	B	kept	
41	Bro018	1387.7	14	119	–	–	B	B	reassigned	a letter covers two participants
42	Bmr024	2428.3	8	82	11	13	B	B	kept	
43	Bro014	2559.0	6	63	0	0	B	B	kept	
44	Bro023	725.6	11	141	47	33	B	B	kept	

ID	Meeting	Start	Seg.	Words	Misplaced	%	Rel.	Reb.	Letters	Note
45	Bro016	732.4	8	88	–	–	A	A	reassigned	a letter covers two participants
46	Bro004	2362.6	7	44	8	18	A	A	kept	
47	Bro028	100.4	8	46	–	–	A	B†	reassigned	a letter covers two participants
48	Bed009	3005.9	22	182	166	91	A	A	kept	
49	Bro016	1752.7	9	67	15	22	B	B	kept	
50	Bro010	303.8	18	88	61	69	B	B	kept	
51	Bro004	1974.2	24	201	136	68	B	B	kept	
52	Bed014	809.3	10	106	–	–	B	–	–	one participant only; dropped
53	Bro027	1958.0	8	58	6	10	A	A	kept	
54	Bro007	1634.5	19	183	153	84	B	B	kept	
55	Bed011	3009.4	10	57	25	44	A	A	kept	
56	Bmr023	3018.5	8	84	37	44	A	A	kept	
57	Bro017	1843.6	14	138	94	68	B	B	kept	
58	Bro005	3498.5	13	132	82	62	A	A	kept	
59	Bro017	490.8	9	113	66	58	A	A	kept	
60	Bmr009	0.0	7	29	1	3	A	A	kept	
61	Bro024	3861.2	13	79	30	38	B	B	kept	
62	Bro005	2377.8	10	49	29	59	B	B	kept	
63	Bro005	1191.3	10	69	44	64	A	A	kept	
64	Bmr009	402.1	16	109	–	–	A	A	reassigned	a letter covers two participants
65	Bmr008	2880.7	10	81	55	68	A	A	kept	
66	Bed004	2208.7	14	140	82	59	B	B	kept	
67	Bro017	1170.9	7	56	–	–	A	A	reassigned	a letter covers two participants
68	Bro023	924.6	11	88	49	56	A	A	kept	
69	Bro004	1866.2	13	75	61	81	A	A	kept	
70	Bro012	2127.7	7	85	1	1	B	B	kept	
71	Bro023	3492.6	14	202	150	74	B	B	kept	
72	Bmr009	429.6	11	55	40	73	A	A	kept	
73	Bed010	2902.1	13	84	55	65	A	A	kept	
74	Bro007	1618.4	7	43	5	12	B	B	kept	
75	Bro026	2907.4	7	95	1	1	A	A	kept	
76	Bro026	193.2	12	55	39	71	A	A	kept	
77	Bro017	1661.9	12	70	41	59	B	B	kept	
78	Bmr006	2068.6	8	57	27	47	A	A	kept	
79	Bed014	2997.4	14	145	–	–	B	A†	reassigned	a letter covers two participants
80	Bmr010	2417.0	8	68	21	31	A	A	kept	
81	Bro017	1031.2	15	214	155	72	B	B	kept	
82	Bed005	2361.9	17	92	72	78	B	B	kept	
83	Bro005	1937.1	8	77	34	44	B	B	kept	
84	Bro017	161.4	16	238	141	59	A	A	kept	
85	Bro008	264.4	9	114	52	46	B	B	kept	
86	Bed008	4062.8	7	46	2	4	B	B	kept	
87	Bro007	8.2	47	197	187	95	A	A	kept	
88	Bro028	3057.3	7	92	–	–	B	A†	reassigned	a letter covers two participants
89	Bro010	170.5	10	93	54	58	A	A	kept	
90	Bmr007	645.4	7	88	1	1	A	A	kept	
91	Bro010	1522.3	12	132	76	58	B	B	kept	
92	Bro018	276.6	8	76	47	62	B	B	kept	
93	Bmr025	219.4	8	75	38	51	A	A	kept	
94	Bed011	1688.3	10	78	47	60	A	A	kept	
95	Bro027	2973.0	13	170	102	60	A	A	kept	
96	Bro004	703.2	24	139	119	86	A	A	kept	
97	Bed004	1947.2	35	295	255	86	B	B	kept	
98	Bro014	2034.0	6	66	0	0	B	B	kept	
99	Bro019	1835.1	8	55	11	20	A	A	kept	
100	Bed002	420.1	12	107	58	54	B	B	kept	
101	Bro004	3620.1	22	93	81	87	B	B	kept	
102	Bmr026	1655.9	9	104	–	–	B	B	reassigned	a letter covers two participants
103	Bmr005	1882.2	21	190	145	76	A	A	kept	
104	Bmr015	2175.0	7	96	21	22	B	B	kept	
105	Bmr024	2201.3	25	127	107	84	B	B	kept	
106	Bro018	2115.4	11	63	42	67	A	A	kept	
107	Bro012	3103.6	8	116	2	2	B	B	kept	
108	Bro016	2392.7	16	73	48	66	B	B	kept	third participant
109	Bro022	2026.3	9	62	38	61	B	B	kept	
110	Bro028	449.3	19	112	103	92	B	B	kept	
111	Bro026	2536.3	12	126	–	–	B	A†	reassigned	a letter covers two participants
112	Bro019	2258.2	16	161	85	53	B	B	kept	

ID	Meeting	Start	Seg.	Words	Misplaced	%	Rel.	Reb.	Letters	Note
113	Bed008	2944.1	13	79	–	–	A	A	reassigned	a letter covers two participants
114	Bro018	705.3	10	65	51	78	A	A	kept	
115	Bro018	2418.2	22	186	136	73	B	B	kept	
116	Bmr007	1128.4	10	83	–	–	A	A	reassigned	a letter covers two participants
117	Bed004	3046.9	17	111	102	92	B	B	kept	
118	Bro012	1095.0	12	86	15	17	B	B	kept	
119	Bro017	2942.0	8	43	–	–	A	B†	reassigned	a letter covers two participants
120	Bro026	248.3	13	93	74	80	B	B	kept	
121	Bro017	2039.3	19	160	–	–	B	A†	reassigned	a letter covers two participants
122	Bro018	1186.9	25	347	331	95	A	A	kept	third participant
123	Bed004	3228.2	10	80	45	56	A	A	kept	
124	Bro028	130.7	7	22	3	14	A	A	kept	
125	Bro024	1896.6	11	96	47	49	B	B	kept	
126	Bmr024	2244.4	24	122	104	85	B	B	kept	
127	Bro016	2058.3	7	30	2	7	A	A	kept	
128	Bmr009	525.8	11	32	–	–	B	B	reassigned	a letter covers two participants
129	Bed010	571.6	9	50	29	58	A	A	kept	
130	Bmr016	107.0	6	51	0	0	A	A	kept	
131	Bro017	530.2	7	37	16	43	A	A	kept	
132	Bmr006	1533.8	11	61	–	–	A	A	reassigned	a letter covers two participants
133	Bro027	3033.0	14	139	70	50	A	A	kept	
134	Bro013	56.7	9	44	24	55	B	B	kept	
135	Bro008	683.9	18	262	236	90	A	A	kept	
136	Bmr006	2774.4	6	73	–	–	A	A	reassigned	a letter covers two participants
137	Bro027	2603.4	15	93	78	84	A	A	kept	
138	Bro017	779.8	11	103	47	46	B	B	kept	
139	Bmr013	1012.2	14	92	72	78	A	A	kept	
140	Bed009	1365.9	9	81	37	46	A	A	kept	
141	Bro019	102.1	16	107	59	55	B	B	kept	
142	Bed005	3404.3	19	144	96	67	A	A	kept	
143	Bro023	1185.4	9	88	39	44	A	A	kept	
144	Bro003	4872.2	9	87	27	31	A	A	kept	
145	Bmr005	4210.8	8	104	43	41	A	A	kept	
146	Bed002	1965.5	11	90	52	58	B	B	kept	
147	Bed009	2960.8	16	102	84	82	A	A	kept	
148	Bro005	1470.7	7	20	5	25	A	A	kept	
149	Bro017	1433.0	6	52	0	0	A	A	kept	
150	Bed008	1505.7	11	44	14	32	A	A	kept	

B Experiments on the released snippets

Table 5 lists what we ran on the 151 released snippets before the rebuilt set existed. Our reimplementation of the XOR graph as an integer program reproduces the published 54.3% exactly (82 of 151). The quote check from our proposal, in which an argument is kept only if a model can quote a supporting span of the snippet verbatim, kept 33.8% of the 7,248 cached arguments and lowered accuracy. A logistic regression fit to the cached validity and grounding scores, evaluated leave-one-out, reaches 53.0 to 57.0%. Rescoring all cached arguments with Jev, including a rebuttal question that shows the rebuttal next to its reason, gives 50.3% under the XOR graph. The first three rows after the baseline, the quote filter and the Gemma 4 run are from our earlier implementation.[5]

C Grounded extension: details

Framework. Each snippet has 48 arguments: 24 reasons, three per speaker per aspect, and one rebuttal for every reason. A reason for X supports the claim that X is the professor. A reason has strength $P(\text{valid}) \cdot P(\text{grounded})$ and a rebuttal has strength $P(\text{counters its reason}) \cdot P(\text{grounded})$. The grounded labelling is the least fixed point of two rules: an argument is IN when every attacker is OUT, so an unattacked argument is IN, and OUT when at least one attacker is IN; any other argument is undecided. The speaker with more IN reasons is predicted, with ties going to the larger sum of reason strengths and then to A.

[5] https://github.com/KrishMay05/nlp_research

Method	Correct	Accuracy (%)
Always A	83	55.0
XOR graph, cached GPT-3.5 scores	82	54.3
Grounded extension, as proposed	69	45.7
XOR graph with quote filter	64	42.4
Gemma 4 zero-shot	66	43.7
Jev, "is X the professor"	72	47.7
XOR graph, Jev scores	76	50.3
Preference-based grounded extension, cached scores	79	52.3
Preference-based grounded extension, Jev scores	71	47.0
Typed-statement graph (Jev)	77	51.0

Table 5: Accuracy on the 151 released snippets. Beating the 55.0% base rate at $p < 0.05$ needs 94 correct.

The proposed relation is degenerate. In our proposal every rebuttal attacks its reason, and every reason for A and every reason for B attack each other. Nothing attacks a rebuttal, so every rebuttal is IN and every reason is OUT. Removing weak rebuttals does not help, because two opposing reasons that attack each other form a cycle, which the grounded labelling leaves undecided. In every run, on released and rebuilt snippets alike, no reason is IN and all decisions fall to the tie-break, that is, to the sum of reason strengths.

The preference relation. Following preference-based argumentation, an attack succeeds only if the attacker is at least as strong as its target: a rebuttal defeats its reason when it is no weaker, and a reason defeats an opposing reason when it is strictly stronger. At most one direction of each attack succeeds, which removes both problems. Table 6 gives the results on the rebuilt snippets.

Arguments	Scores	Relation	Accuracy (%)	Decided by IN count	Reasons IN
Terra	Jev	proposed	66.7	0 of 150	0 of 3,600
Terra	Jev	preference	70.0	150 of 150	1,504 of 3,600
GLM	Jev	proposed	71.3	0 of 150	0 of 3,600
GLM	Jev	preference	67.3	150 of 150	1,438 of 3,600
Haiku	Jev	proposed	73.3	0 of 150	0 of 3,600
Haiku	Jev	preference	73.3	150 of 150	1,517 of 3,600
Haiku	Haiku	proposed	68.7	0 of 150	0 of 3,600
Haiku	Haiku	preference	64.7	141 of 150	849 of 3,600

Table 6: Grounded extension on the 150 rebuilt snippets at the original span.

Policy search. To ask whether some other score-aware policy does better, we evaluated 288 policies over the Haiku arguments and Jev scores. They vary the strength of a reason (three definitions) and of a rebuttal (three definitions), the rule for when a rebuttal attacks (preference, or a threshold of 0.3, 0.5 or 0.7 on its strength), the cross-side rule (none, stronger reason defeats any opposing reason, the same within an aspect only, or only between reasons Jev judges similar), and the decision rule (count or summed strength of IN reasons). Meetings were split at random into a development half and a test half; the best development policy was scored on the test half next to the direct Jev question on the same half. Over 500 splits the selected policy averages 69.7% and the direct question 71.5%; the selected policy is ahead in 133 of 500 splits (mean difference -1.8 points, middle 95% of splits -11.2 to $+3.8$). On all 150 snippets the policies range from 82 to 111 correct with a median of 99, against 107 for the direct question. The same search over GLM's arguments gives 67.1% for the selected policy against 71.5%, ahead in 41 of 500 splits, with policies ranging from 89 to 108 correct.